

FOX_Checklist

What this is: every FoxByte feature and its sub-features — built or not, and how it was tested — so it's clear where we are and what's next. How each feature works is in `FOX_Feature_Implemented.pdf` (same ids).

Last updated: 21 September 2026 · **Reflects:** `main @ 2778762` (v1.0.1, exact repository path PR #1) + the FoxByte rename and brand indirection (`chore/foxbyte`)

Source: `docs/FOX_Checklist.html` → `make feature-doc` → this PDF

✔ Done & tested — automated tests (unit, VM integration or CI)

🔸 Partial — gaps, or not verified end to end

Sub-features: ✔ built 🔸 partly ☐ not built yet

`make integration-v2` ; "update" = `make integration-update`

❌ Known issue

☐ Not started

❌ broken · "VM §n" = section of `make integration` ; "v2 §n" =

Where we are

Area	Features	Done & tested	Done (manual)	Partial	Known issue	Not started	Progress
A · Install & setup	9	4	3	2	·	·	78%
B · CLI & services	2	2	·	·	·	·	100%
C · Branching & storage	8	6	1	1	·	·	88%
D · Durability	4	3	·	1	·	·	75%
E · Security foundations	3	3	·	·	·	·	100%
F · SQL gateway	7	6	1	·	·	·	100%
G · Accounts & auth	5	4	·	1	·	·	80%
H · Blackbox (record of schema changes)	18	17	·	1	·	·	94%
I · Import & migration	13	5	8	·	·	·	100%
J · ETL pipelines (experimental)	4	2	2	·	·	·	100%
K · APIs, SDKs & agents	6	4	2	·	·	·	100%
L · Web console	11	1	10	·	·	·	100%
M · Build, CI & releases	6	4	2	·	·	·	100%
Total (features built or started)	96	61	29	6	·	·	94%
N · Not started (roadmap)	11	·	·	·	·	11	0%

Storage engine — not implemented, by design

FoxByte does **not** have its own storage engine. Data is stored by **PostgreSQL 16's own engine** (heap tables, B-tree indexes, write-ahead log) running on local disk inside the stock `postgres:16-bookworm` image plus wal-g. This was a locked architecture decision: a Neon-style engine that separates compute from storage over the network adds latency to every transaction, and transaction speed must never be compromised.

What **is** built is a storage *layer* around Postgres: copy-on-write branch drivers — ZFS on macOS/Linux, btrfs on Windows — behind one interface in `internal/branch/storage.go` (C1), quotas (C6), capacity reporting (C7), and durability to object storage with wal-g + MinIO: WAL archiving, base backups, point-in-time restore (D1–D3). Networked storage disaggregation is listed in the README roadmap but not started (see "Not started").

Suggested next steps

- Run `fox update` against a real release on macOS and Windows (A9) — cut v0.9.0, install it, then update to v0.9.1.
- HA beyond the demo: archive WAL while main runs on the promoted standby, then automatic failover and fencing (D4). (*Safety fixes C3, D4 hazards, H7, I5, I6 done.*)
- Verify downloads everywhere: installers and `fox setup`'s engine download still skip `SHA256SUMS` (A1, A2, A7).

4. ~~Check the web pages that were built but never used signed in: Integrity, Policies, sidebar (L8–L10); add Blackbox filters (L5).~~ Done — both, in the browser.
5. Add automated tests where there are none: imports (I1–I3, I7 — I5 and I8 now covered), SDKs (K3, K4), web UI (L*).
6. Run the VM integration suites in CI (M2).

A Install & setup

7 of 9 done

ID	Feature · sub-features	Tested by
 A1	One-line installer (macOS / Linux) ✓ Detects OS / CPU, downloads the release binary ✓ Installs to the prefix (sudo only if needed) ✓ Stages the Linux engine for the VM on macOS	Unit (<code>test_install_sh.sh</code> , CI); manual
 A2	Windows installer (install.ps1) ✓ Installs WSL, elevates, resumes after reboot ✓ Downloads bb.exe, engine, image build context ✓ Ubuntu rootfs verified against Ubuntu's checksums ✓ Adds to PATH and runs <code>fox setup</code>	Pester (Windows CI); Windows run 16 Sep
 A3	fox setup on macOS (Lima) ✓ Creates / starts the Lima VM ✓ Installs Docker and ZFS ✓ Reinstalls the engine on every run ✓ Starts the stack	Manual (from-scratch VM run)
 A4	fox setup on Windows (WSL2 + btrfs) ✓ Imports a dedicated distro with systemd ✓ Docker, btrfs and a storage mount unit ✓ Quiet steps, detail in install.log ✓ Waits for systemd on a cold distro	Unit (<code>host_wsl_test.go</code>); Windows run 16 Sep
 A5	Linux native ✓ Engine runs in-process, no VM	VM integration (all suites run on Linux)
 A6	Command forwarding & host file streaming ✓ macOS → Lima VM, starts the VM if stopped ✓ Windows → WSL distro	Unit (<code>TestImportLocalFile</code> , <code>TestWSLArgs</code>)
 A7	Always-latest engine binary ✓ Downloads the newest engine into the cache (ELF check) ✓ <code>FOX_NO_REFRESH</code> , <code>FOX_VERSION</code>	Unit (<code>host_test.go</code> , <code>update_test.go</code>); Windows run 16 Sep
 A8	New-release notice on fox start ✓ Background check, 1.5 s limit, printed last ✓ Only a promoted, newer, complete release	Unit; update \$1, \$1b, \$2, \$3, \$8; real Mac check
 A9	fox update ✓ Downloads and verifies everything before changing anything ✓ Stops servers, installs engine, restarts old one on failure ✓ <code>up</code> , Blackbox upgrade, <code>start</code> on the new engine ✓ Replaces host <code>fox</code> , keeps previous binaries	Unit; update \$4–\$7; a real Mac run

B CLI & services

2 of 2 done

ID	Feature · sub-features	Tested by
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✓	B1 The fox command set	✓ Stack: setup, start, stop, up, down, status, logs, psql, update ✓ Branching, backup / restore, HA ✓ Blackbox, policy, impact ✓ Import, pipelines ✓ Users, API keys, admin	✓ <code>fox vm status / fox vm shell</code> (Lima, WSL; Linux says there is no VM) □ <code>branch create --from</code> ✓ <code>fox uninstall</code> removes it all, retired installs included	Unit (<code>main_test.go</code> , <code>update_test.go</code>); VM suites
✓	B2 Background services (daemon)	✓ Control plane :8080, gateway :6432, agent API :8088 ✓ Detached, pidfiles and logs in <code>~/ .fox</code>	✓ Stop (TERM then KILL), status, logs □ Health check / restart on crash	Unit (<code>TestStopTerminatesProcess</code>); VM §1

C Branching & storage

7 of 8 done

ID	Feature · sub-features	Tested by
✓ C1	Storage drivers (ZFS / btrfs) ✓ ZFS snapshot + clone per branch (macOS VM, Linux) ✓ btrfs subvolume snapshots (Windows) ✓ Pool / filesystem created on a loopback file	Unit; VM §2, §2b (ZFS); btrfs manual on Windows
✓ C2	Branch lifecycle ✓ Create, list, delete ✓ Reset to parent ✓ Suspend / resume ✓ Recreate a container from its surviving data	Unit; VM §2, §2b, §4
✓ C3	Protection of main ✓ Delete and reset refused	Unit; VM §4, §6
✓ C4	Private networking (no host ports) ✓ Branch databases reachable only through the gateway	Unit (<code>TestParsePublishedPort</code> , <code>TestDSN</code>)
✓ C5	Scale-to-zero & wake on connect ✓ Idle branches suspended automatically ✓ Next connection wakes the branch (~0.3–0.7 s)	VM §7
🔗 C6	Storage quotas ✓ Space reserved for main (ZFS) 🔗 Per-branch cap (opt-in)	Manual
✓ C7	Capacity & status ✓ Services, main health, backups, branches ✓ Storage used / available	VM §1
✓ C8	Provisioning & image pull ✓ Docker, storage pool and image prepared on up	Manual

D Durability

3 of 4 done

ID	Feature · sub-features	Tested by
✓ D1	WAL archiving to object storage ✓ wal-g ships main's WAL to MinIO, off the commit path	VM §3
✓ D2	Base backups ✓ Create and list ✓ Follow the primary after a failover; listing works with main stopped	VM §3, §3b, §6

✓	D3 Point-in-time restore	✓ Restore to a timestamp or latest on :5433 ✓ Per-install credentials ✓ Starts from the newest base backup that precedes the target	✓ A point no backup precedes is refused, naming the oldest	Unit; VM §3, §3b, §6; v2 §1a
🔥	D4 HA standby & failover	✓ Streaming hot standby ✓ Manual failover, <code>main</code> rerouted ✓ Nothing deletes or stops the promoted standby ✓ <code>fox up</code> brings it back, not the old main	✓ <code>fox ha failback</code>, keeping every write ✓ The promoted standby archives WAL, so backups and PITR cover the failover ❑ Automatic failover, fencing, multi-host	Unit; VM §6, §7

E Security foundations 3 of 3 done

ID	Feature · sub-features	Tested by
✓ E1 Per-install generated secrets	✓ Generated Postgres and MinIO passwords ✓ Used by restore too	Unit (indirect); v2 §1a
✓ E2 TLS everywhere	✓ Self-signed TLS on all three services, zero setup 🔥 TLS optional at the gateway (<code>sslmode=disable</code> still connects)	Unit (<code>TestReadStartupUpgradesTLS</code>)
✓ E3 Keys are made by a person, not by start	✓ Start creates no account and no key, and caches none on disk ✓ The banner shows the connection string and how to make a key ✓ The first account created gets the admin grant on main	Integration §1

F SQL gateway 7 of 7 done

ID	Feature · sub-features	Tested by
✓ F1 Routing by database name	✓ One endpoint :6432, database name = branch ❑ Connection pooling, rate limiting	Unit; VM §1, §6
✓ F2 API-key authentication	✓ <code>key_</code> key as the Postgres password ✓ Bad key rejected with a Postgres error	VM §1
✓ F3 Backend login (SCRAM)	✓ Cleartext, md5, SCRAM-SHA-256 ✓ Clients never see the database password ❑ Unit test for SCRAM	VM §1, §6, §8 (exercised)
✓ F4 Per-user Postgres login roles	✓ Role per user, actor can't be forged ✓ Permissions of non-superuser <code>db_client</code>	Manual
✓ F5 Attribution session settings	✓ Actor kind, branch, session id per connection 🔥 Kind / branch / session client-settable	VM §8
✓ F6 Ctrl-C (CancelRequest) forwarding	✓ Cancel a running query through the gateway	Unit
✓ F7 Errors & insecure build tag	✓ Real Postgres error responses ✓ No-auth bypass impossible in release builds	Unit (<code>TestSendError</code>)

G Accounts & auth 4 of 5 done

ID	Feature · sub-features	Tested by
✓ G1 Auth store (SQLite)	✓ Users, keys, sessions, OAuth ids, pipelines ✓ Safe under concurrent use (WAL, busy timeout), including a new store created by three services at once ❑ Roles / permissions, multi-tenant projects	Unit (<code>TestVerifyKeyConcurrent</code> , <code>TestConcurrentFirstOpen</code>); VM §1

✓	G2 Email & password users ✓ Create account, log in □ Password change / reset	□ List and delete users	Unit; VM §1
✓	G3 API keys ✓ Create (shown once), list, revoke ✓ Bearer / X-API-Key and gateway password	○ Revoking an unknown id reports success	Unit; VM §1
✓	G4 Web sessions & signup control ✓ httpOnly session cookie	✓ Signup can be closed	Unit (TestSessions)
○	G5 OAuth (GitHub / Google) ✓ Sign-in flows built ✓ Defaults match what the engine serves (https://localhost:8080 ; web origin = public URL)	✓ Session cookie Secure / SameSite follow the origin □ Tested with a real OAuth app	Unit; VM §1b

H Blackbox (record of schema changes)

17 of 18 done

ID	Feature · sub-features	Tested by
✓	H1 DDL recording ✓ Who, tool, branch, statement, status, risk ✓ Inherited by every branch	□ Role / database changes, some dropped object types VM §8
○	H2 Hash chain, append-only & verify ✓ Rows can't be edited or deleted normally ✓ fox ledger verify ○ Superuser rewrite caught only via anchors (H10–H12)	✓ bb.policy changes recorded (append-only, not chained) VM §8; v2 §0, §6b
✓	H3 Attribution ✓ Humans via their login role ✓ Agents via their own agent-<id> role	✓ Kind derived from the role, so neither can pose as the other ✓ Tool and session are client-declared by design, and labelled so Unit; VM §8; v2 §1c
✓	H4 Destructive-DDL guardrail ✓ DROP TABLE / DROP SCHEMA refused, attempt recorded ✓ Override only superusers / db_admin , use recorded	✓ "flag" and "allow" actions; unknown actions refused ✓ No hang when blocked after a write in the same transaction VM §8; v2 §1b, §6b
✓	H5 Risk flags ✓ Type change, drop column flagged ✓ Judged on the running statement (no comments or literals)	✓ Automated test Unit; v2 §6b
✓	H6 Blackbox access (CLI / REST / MCP / UI / SDK) ✓ CLI, REST with filters, MCP, web page, SDKs	□ Per-user access control v2 §0, §5
✓	H7 fox ledger revert --to ✓ Help and output say it's a point-in-time restore of main	✓ Points to H13 for a branch from before a change Unit (help text); VM §3 (the restore)
✓	H8 Override permission (db_admin) & upgrade ✓ fox admin grant revoke list ✓ First user granted automatically	✓ fox blackbox upgrade [--all] ✓ Grant / revoke from the web (admins only) v2 §1b, §6
✓	H9 Blackbox 2.0 capture ✓ Transaction id and WAL position per entry ✓ Task, tool-call hash, override use	✓ Fail-safe, append-only ✓ Kill switch not settable by clients Unit; v2 §2
✓	H10 Checkpoints & anchors ✓ Merkle root in the database and in read-only anchor files ✓ On a schedule and on demand	✓ Published anchor format □ Cloud write-once anchors (S3 Object Lock) Unit; v2 §3, §3b
✓	H11 fox-verify (independent verifier) ✓ Checks a live branch or an export	✓ Release binaries for every platform Unit; v2 §3

✓	H12 Blackbox integrity & export	✓ INTACT / TAMPERED via CLI, REST, MCP ✓ JSON-lines export	❑ Streaming for very large ledgers	v2 §3
✓	H13 Branch from just before a Blackbox entry	✓ Restore + replay up to the change's transaction ✓ By transaction id or timestamp	✓ CLI, REST, MCP, web panel ✓ main untouched	Unit; v2 §4
✓	H14 Blackbox names & aliases	✓ <code>fox blackbox</code>, <code>/blackbox</code> routes, MCP and SDK names	✓ Every original name still works	Unit; v2 §5
✓	H15 Policy gate (warn / block)	✓ BBX02 warn / BBX01 block with JSON detail ✓ Blocked attempts recorded ✓ Per-rule <code>db_admin</code> override	✓ CLI, REST, MCP, web page; custom rules ✓ Patterns match the running statement ❑ TRUNCATE	Unit; v2 §6, §6b
✓	H16 Agent provenance & execute_change	✓ Session, task, parent session on every agent change (also through the Gateway DSN) ✓ MCP <code>execute_change</code> with policy preview	✓ <code>fox blackbox sessions</code>, REST ⚠ Provenance is what the agent says (not verified)	Unit; v2 §7
✓	H17 Impact analysis	✓ Dependents (views, foreign keys, indexes, depth) ✓ Other branches that have the object ✓ Policy verdict and risk score	✓ CLI, REST, MCP, inside <code>execute_change</code> ⚠ Functions, types, schemas not recognised from SQL	Unit; v2 §8
✓	H18 Blackbox diff between branches	✓ Fork point, changes only on each side ✓ Objects changed on both	✓ CLI, REST, MCP ❑ Merge (diff only today)	Unit; v2 §8

I Import & migration

13 of 13 done

ID	Feature · sub-features	Tested by
✓ I1	From Postgres ✓ Schema and data via <code>pg_dump</code> ✓ Zero-table result reported as an error	Manual
✓ I2	From MySQL ≤5.7 / MariaDB ✓ pgloader with type mapping (multi-arch image)	Manual
✓ I3	From MySQL 8.x ✓ Native mysqldump path (<code>cached_sha2_auth</code>) ✓ Types, NULLs, unicode, PK	Manual
✓ I4	From MongoDB ✓ Collection → typed table, exact name	VM §9, §10
✓ I5	From a .sql dump ✓ Runs the dump into a new branch ✓ Fails listing every failed statement (instance kept)	Unit; VM §11
✓ I6	From JSON / NDJSON ✓ Columns inferred from top-level keys	Unit; VM §10, §11
✓ I7	From CSV ✓ Table from the header, exact column names	Manual

✓	I8	Continuous replication & cutover ✓ Initial copy + streaming (logical replication) ✓ Cutover keeps data (CLI) ✓ Status and cutover from REST and the Import page ✓ A replicating branch is never auto-suspended ✓ Automated test (VM §5b) and driven in the browser	Unit; VM §5b
✓	I9	Guard handling during import ✓ Guardrail paused, recording stays on ✓ Re-armed even after a crash	Manual
✓	I10	Live progress (CLI & SSE) ✓ Live log and progress bar ⚠️ psql subprocess output not streamed	Manual
✓	I11	Import REST endpoints ✓ From a connection string or an uploaded file ☐ Target name validation	Manual
✓	I12	Web Import page ✓ Database and file panels with live console ⚠️ Mentions a SQLite loader that doesn't exist	Manual
✓	I13	Exact identifier fidelity ✓ Case kept (<code>userId</code> , <code>leaseAiChats</code>) ⚠️ Field order of JSON sources not kept	Unit; VM §10

J ETL pipelines (experimental) 4 of 4 done

ID	Feature · sub-features	Tested by
✓	J1 Pipeline engine ✓ Extract → raw schema → SQL models → tests, in a fresh branch ✓ <code>ref()</code> / <code>source()</code> templating ☐ Dependency graph ☐ Incremental models ☐ Scheduling	Unit; VM §9
✓	J2 Data-quality tests ✓ <code>not_null</code> , <code>unique</code> , <code>accepted_values</code> , <code>row_count_min</code> , <code>custom</code> ✓ Failure → exit 1, data kept	Unit; VM §9
✓	J3 Stored pipelines & run history ✓ Saved per user, run over the API, history ☐ Read a run's log ☐ Mark runs whose process died	Manual
✓	J4 Pipeline editor (web) ✓ Create, edit, run, results, history ☐ Confirm before delete	Manual

K APIs, SDKs & agents 6 of 6 done

ID	Feature · sub-features	Tested by
✓	K1 Control-plane REST API ✓ Branches, query, Blackbox, policy, impact, import, pipelines, keys ☐ Versioned paths (<code>/api/v1</code>)	Unit; VM §1; v2 §0–§8
✓	K2 OpenAPI spec ✓ Served, Blackbox 2.0 endpoints documented ✓ Matches the code (import fields, status codes, auth routes, <code>/blackbox</code> aliases) ✓ Drift check in CI	Unit (drift check)
✓	K3 Python SDK ✓ Branches, query, Blackbox ☐ Import, pipelines, keys, agents ☐ Tests ☐ Publish to PyPI	Manual
✓	K4 TypeScript SDK ✓ Same client for Node and browsers ☐ Tests ☐ Publish to npm	Manual
✓	K5 Agent Branch API ✓ A branch per agent, own non-superuser role ✓ Provenance (task, parent, session) ✓ DSN reachable from the host (gateway-routed, branch-scoped key) ✓ Cap, listing and TTL count suspended branches	Unit; VM §5; v2 §1c, §7

✓	K6 MCP server	Unit; v2 §1c–§8
✓	- Branches, run_sql, Blackbox tools - policy_check, execute_change, impact, diff - Requires an API key and acts as that account; a scoped key is held to its branch	Client config & 16-tool reference (docs/mcp.md)

L

Web console

11 of 11 done

ID	Feature · sub-features	Tested by
✓ L1	Embedded single-origin UI ✓ Served by the engine at :8080 □ Automated UI tests (whole area)	Manual
✓ L2	Login & signup ✓ Log in / sign up tabs, OAuth buttons	Manual
✓ L3	Dashboard ✓ Live tiles ✓ Branch table: create, suspend, resume, delete, copy DSN	Manual
✓ L4	SQL console & browser ✓ Schema sidebar, rows, structure, indexes ✓ Values shown as Postgres writes them (uuid, interval, inet, bytea) ✓ SQL editor ✓ System tables hidden by default, with a show button ✓ Runs as the signed-in user; one-run override switch ✓ Per-rule override for policy blocks	Manual; API in v2 §6
✓ L5	Blackbox page ✓ Entries per branch with badges ✓ Table / risk / date filters, and "Clear filters" ✓ Verify button; a row expands to its statement and session	v2 §5b; browser
✓ L6	API keys page ✓ Create, copy, list, revoke ✓ A new key is shown inside its connection string (E3's next step)	Manual; browser review 21 Sep
✓ L7	Landing, Guide & Docs pages ✓ Landing, developer guide, reference ✓ Refreshed against the code (https, sslmode, MCP, Blackbox, import routes)	Type-check + build
✓ L8	Blackbox integrity page ✓ Verify, checkpoint, export, branch-before ✓ Used signed in; every result matched the CLI ○ Entry list needs a refresh to show a new change	Type-check + build; browser review 16 Sep
✓ L9	Policies page ✓ Rules, warn/block, custom rules, preview, evaluations ✓ Used signed in; matched the CLI, wrote only to the chosen branch ✓ Non-admin change refused with a readable message ✓ Who can override: list, grant, revoke	Type-check + build; API in v2 §6; browser review 16 Sep
✓ L10	App sidebar layout ✓ Every link opens its page, highlighted, with breadcrumb ✓ Collapse persists across a reload; theme toggle persists ✓ No console errors on any page □ Under-860px drawer (needs a manual look)	Type-check + build; browser review 16 Sep
✓ L11	Design system & FoxByte theme ✓ Tokens for colour, type, space, radius, elevation and motion; components use no raw values ✓ Ember palette in both themes; filled controls legible (--accent-ink) ✓ Fox mark from theme variables, wordmark from brand.json, favicon ✓ Console, Blackbox, Policies and API keys re-laid out; one table style ✓ Every page reviewed in both themes; console checked at 375px □ Not yet reviewed against a running engine with real data	Type-check + build; browser review 21 Sep

M

Build, CI & releases

6 of 6 done

ID	Feature · sub-features	Tested by
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✓	M1 Makefile targets	✓ build, vet, test, integration suites ✓ Suites run only in a throwaway test VM (guarded)	✓ release, web, WSL builds, feature-doc (both PDFs)	CI
✓	M2 Linux CI	✓ Build, vet, test, gofmt ✓ Release script tests, checklist sync check	□ VM integration suites in CI	CI
✓	M3 Windows CI	✓ Build, vet, test, Pester, installer encoding	□ End-to-end WSL install test	CI
✓	M4 Releases & GHCR image	✓ Binaries for 5 platforms + fox-verify ✓ SHA256SUMS merged safely between jobs	✓ Fails unless fox update 's files are published ✓ Multi-arch Postgres image on GHCR	Script tests in CI
✓	M5 WSL distro & ZFS builds	✓ Prebuilt distro image ✓ Downloaded and verified by the installer (seen on Windows) ✓ Preloaded image tagged as the engine runs it	✓ Imported on a fresh Windows install; preloaded images loaded 🟡 685 MB download (rootfs fallback covers failures)	Actions artifacts; Pester; Windows run 16 Sep
✓	M6 Postgres image	✓ Postgres 16 + wal-g	□ pgvector	Manual

Planned or discussed, nothing built yet. When work starts, move the row into its area with the next id there.

ID	Item · scope	Related
☐ -	Multi-tenant projects & roles ☐ Projects / organisations ☐ Per-branch permissions (RBAC)	G1, H6
☐ -	Branch merge ☐ Apply one branch's schema changes to another	H18
☐ -	Scheduled backups & retention	D2
☐ -	Automatic HA failover ☐ Health detection, fencing ☐ Multi-host	D4
☐ -	Pipelines: scheduling, incremental / CDC extract, DAG view	J1
☐ -	More import sources: Cassandra, DynamoDB	I1–I8
☐ -	pgvector in the Postgres image	M6
☐ -	Cloud write-once anchors (S3 Object Lock etc.)	H10
☐ -	Release signatures	A9, M4
☐ -	Hosted cloud tier	—
☐ -	Networked storage disaggregation (README roadmap; see the storage engine note on page 1)	C1, D1

Out of scope by decision

A custom storage engine (PostgreSQL's own engine is used, for native transaction speed), sharding, Raft / distributed transactions, multi-region, HTAP.